

Nishal Stanislaus Silva, PhD
Toronto, ON
nishal.silva@hotmail.com | +1 613 200 2030
nishal.xyz | linkedin.com/in/nishal-silva | github.com/ns2max

Dear Hiring Manager,

I am writing to apply for the Engineer, AI Software/Machine Learning (C++) role with Qualcomm's Machine Learning Engineering group in Markham. The posting is explicit that this is a C++ engineering role, not model building or training, and that is exactly the work I have spent my career doing: writing modern C++17 software that makes trained models run fast and reliably on constrained hardware. My own open-source project, Nebula, is a C++17 audio-feature library with per-feature latency benchmarks on ARM, and my published research detector runs at 14 ms on a Raspberry Pi 4, 74x faster than a DTW baseline, at F1 0.76 (JAES 2026).

Against the specifics of the posting: I have built and shipped C++/Python real-time inference engines on embedded and IoT hardware across manufacturing sites for MAS Holdings, including reverse-engineering PLC trigger points and building three closed-loop automation subsystems in custom C++, and I design frozen-protocol benchmark and ablation suites as part of my research practice, the same discipline a model-analysis tool for customers requires. My embedded-toolchain experience is concentrated on Raspberry Pi and embedded Linux (Elk Audio OS) rather than Android or QNX specifically, and I have not worked directly with the Hexagon DSP SDK; both are a natural extension of the cross-platform, memory- and latency-constrained C++ work I already do, and I would expect to be productive on them quickly. I use TensorFlow, PyTorch and ONNX routinely as the consumer of the models my tooling has to serve, and I work daily in Git and Docker.

Qualcomm's Hexagon-based on-device AI stack is the kind of hard compute-and-power-budget problem I have chosen to specialize in rather than one I am pivoting toward, and the Markham site puts that work close to home. I would be glad to bring my benchmark-driven approach to latency and CPU trade-offs to the team's model-analysis tooling from day one.

I am a Canadian Permanent Resident and do not require sponsorship. I am based in Toronto, a short commute from Markham, and available immediately. I would welcome the chance to discuss how my embedded C++ and ML-systems background can contribute to your team.

Sincerely,
Nishal Stanislaus Silva